

Assignment 03	Hepker, Sprout
	2/6/2025

COPY Table 7.1 Cost-Benefit Analysis of Proposed Health Warnings: Central Case (\$m 2003 prices) (a)

	2004	'05	'06	'07	'08	'09	'10	'11	'12	'13	'14	'15	'16	'17	'18	'19	'20	'21	'22	'23	'24	'25	'26	'27	'28	'29	'30
Costs to tobacco industry																											
Printing costs – cigarettes (b)	16	0	0	0	0	0	0	0	0	0	0	16	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Printing costs – cigars/other (b)	2	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Loss of gross profits (cigarettes) (c)																											
Tobacco growers	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Manufacturers	8	19	19	19	19	18	18	18	17	17	17	16	16	16	16	15	15	15	15	15	14	14	14	13	13	13	12
Retailers	4	9	9	9	9	9	8	9	8	8	8	8	8	8	8	7	7	7	7	7	7	7	6	6	6	6	6
Total	30	29	29	29	29	28	27	28	26	26	26	43	25	25	25	23	23	23	23	23	22	22	22	20	20	20	19
Gross costs to government																											
Information programs	5	5	0	0	0	0	5	0	0	0	0	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Loss of excise/customs revenue	70	139	137	135	133	131	129	128	126	124	122	120	118	116	114	112	110	108	106	104	102	100	98	96	94	92	90
Total	75	144	137	135	133	131	134	128	126	124	122	125	118	116	114	112	110	108	106	104	102	100	98	96	94	92	90
Total costs	105	173	167	164	162	159	161	155	152	150	148	168	143	141	139	135	133	131	129	127	124	122	120	116	114	112	109
Benefits																											
Value of increased length of life	0	100	206	215	225	235	245	257	262	267	272	277	282	287	292	297	301	304	304	304	304	304	304	304	304	304	304
Value of increased quality of life	0	30	62	65	68	71	74	77	79	80	82	83	85	86	88	89	90	91	91	91	91	91	91	91	91	91	91
Health care savings government	0	6	13	13	13	14	14	14	14	15	15	16	16	17	17	18	18	19	19	19	19	19	19	19	19	19	19
Gains to non-tobacco firms	0	16	16	16	16	15	15	15	15	15	14	14	14	14	13	13	13	13	12	12	12	12	11	11	11	11	11
Total benefits	0	152	298	309	322	335	348	364	370	377	383	390	397	403	410	417	422	427	426	426	426	426	425	425	425	425	425
Net benefits	-105	-21	131	145	160	176	187	209	218	227	235	222	254	262	271	282	289	295	297	299	302	304	305	309	311	313	316
t	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27
PV Factor	0.952	0.907	0.864	0.823	0.784	0.746	0.711	0.677	0.645	0.614	0.585	0.557	0.530	0.505	0.481	0.458	0.436	0.416	0.396	0.377	0.359	0.342	0.326	0.310	0.295	0.281	0.268
PV (Benefits) QoI Omitted	0.00	110.66	203.00	200.74	199.02	197.00	194.73	193.58	187.58	182.33	175.99	170.95	165.46	160.11	154.89	150.26	144.85	139.61	132.57	126.26	120.25	114.52	108.74	103.56	98.63	93.93	89.46
PV (Benefits) LoI Omitted	0.00	47.17	78.61	77.33	76.00	74.62	73.20	71.74	69.62	67.53	64.90	62.92	60.99	58.59	56.76	54.97	52.79	51.11	48.28	45.98	43.79	41.71	39.39	37.52	35.73	34.03	32.41
PV (Benefits) Combined	0	137.87	257.42	254.22	252.30	249.98	247.32	246.37	238.51	231.45	223.93	217.17	210.54	203.54	197.22	191.03	184.12	177.43	168.58	160.55	152.91	145.63	138.37	131.78	125.50	119.53	113.84
PV (Costs)	100	156.92	144.26	134.92	126.93	118.65	114.42	104.91	97.98	92.09	86.53	93.55	75.84	71.21	66.86	61.85	58.03	54.43	51.05	47.86	44.51	41.71	39.07	35.97	33.66	31.50	29.20

Table 1				Quality of life benefits omitted				Length of life benefits omitted			
Combined Benefits				Discount Rate				Discount Rate			
Discount Rate				1.05				1.05			
PV (Benefits)				4977.08				3918.67			
PV(Costs)				2113.90				2113.90			
NPV				2863.18				1804.77			
BCR				2.35				1.85			
								NPV			
								-656.20			
								BCR			
								0.69			

Table 2																	Average
	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	
Deaths Avoided (Interpolated)	332	348	365	381	398	414	421	429	436	444	451	458	466	473	481	488	
Benefits Value \$m (calculated)	261	274	287	300	313	326	332	338	343	350	355	361	367	372	379	384	
Sum of Values \$m (from Table 7.1)	268	280	293	306	319	334	341	347	354	360	367	373	380	386	391	395	
Amount Overstated \$m	7	6	6	6	6	8	9	9	11	10	12	12	13	14	12	11	9

Table 3																	Averages
	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	
Sum of Values \$m (from Table 7.1)	268	280	293	306	319	334	341	347	354	360	367	373	380	386	391	395	
Deaths Avoided (Table 5.2)	332	-	-	-	-	414	-	-	-	360	-	-	-	-	-	-	
Deaths Avoided (Interpolated)	332	348	365	381	398	414	421	429	436	444	451	458	466	473	481	488	
(Values x 1m) / Deaths Avoided (int)	807228.92	803673.94	803179.82	802728.23	802313.88	806763.29	809207.40	809235.07	811554.33	811541.93	813747.23	813699.83	815800.77	815722.74	813566.38	809426.23	

Table 4																	Averages
	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	
divide by 9 years gained (should be 87,500)	89692.10	89297.10	89242.20	89192.03	89145.99	89640.37	89911.93	89915.01	90172.70	90171.33	90416.36	90411.09	90644.53	90635.86	90396.26	89936.25	89926.32
divide by DALY value of 87,500 (should be 9)	9.23	9.18	9.18	9.17	9.17	9.22	9.25	9.25	9.27	9.27	9.30	9.30	9.32	9.32	9.30	9.25	9.25

Table 5																	Averages
	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	
Divide stated benefits by 9 & 87,500 (should = dths avoided)	340	356	372	389	405	424	433	441	450	457	466	474	483	490	497	502	
Deaths Avoided (Interpolated)	332	348	365	381	398	414	421	429	436	444	451	458	466	473	481	488	
Amount Overstated	8	7	7	7	7	10	12	12	13	14	15	15	17	17	16	14	12